

TECHNICAL BRIEFING & PORTFOLIO

Enterprise AI Engineering for Tier-1 Banking

Production LLM Architectures, RAG Systems, Multi-Agent Workflows, and Regulated AI Governance

Candidate: Rifat **Role:** AI Engineer **Client:** TCS / Morgan Stanley **Clearance:** UK SC & NATO

Battle-Tested Enterprise AI Delivery



Investment Banking

Production delivery across
Goldman Sachs environments.



RAG at Scale

Systems indexing over 10M
enterprise documents.



Governance & Clearance

Active UK SC and NATO security
clearances.

Enterprise RAG System Architecture

Hybrid Search Retrieval

Dense semantic vectors paired with BM25 keyword matching.

Context-Aware Chunking

Hierarchical markdown and semantic document splitting.

Re-Ranking & Filtration

Cross-encoder scoring prioritizing precision over recall.

Vector Database Stack

Proven production selection criteria based on latency, indexing throughput, and enterprise RBAC:

Pinecone & Qdrant: High-concurrency managed retrieval

Weaviate & Milvus: Multi-tenant banking scale

FAISS & ChromaDB: Low-latency local embedded caching

MEASURED PRODUCTION BENCHMARKS

Quantified Performance & ROI

–60%

Retrieval Latency

Optimised vector search
pipelines.

+40%

Context Precision

Domain-specific
chunking strategies.

300%

Deployment Rate

Goldman Sachs CI/CD
agentic framework.

–50%

Token API Costs

Aggressive semantic
caching layers.

Multi-Agent Orchestration & Tool Calling

LangGraph & CrewAI

Cyclic state graphs with checkpoint persistence.

Deterministic routing

ReAct & Tool Use

Schema-enforced function calling patterns.

Zero hallucinated inputs

MCP Integration

Model Context Protocol tool execution.

Decoupled integrations

GitHub Copilot & Claude Code Practice

GitHub Copilot Enterprise

Custom Instructions: Standardised repository
`.github/copilot-instructions`.

Test Synthesis: Automated unit and regression test
scaffolding.

Code Review Agents: Pre-commit static security
analysis.

Claude Code & Mentorship

Terminal Agentic: Autonomous multi-file refactoring
workflows.

Enterprise Enablement: Structured developer
onboarding workshops.

Governance Rules: Safe coding practices preventing
secret leakage.

Rigorous Prompt & Model Governance

RAGAS Framework

Faithfulness and answer relevancy tracking.

Zero regression deployments

LangSmith Telemetry

End-to-end trace latency and token cost audit.

Full audit trail

LoRA / QLoRA Tuning

Parameter-efficient adaptation on private data.

35% cost optimization

AI Security in Regulated Finance



Injection Defense

95% precision prompt guardrails.



PII Redaction

Automated regex and NER sanitization.



Vetted Clearance

UK SC (2028) and NATO (2029).

Morgan Stanley Delivery Roadmap

DAYS 1–30

Discovery & Guardrails

Audit existing repos, models, and compliance boundaries.

DAYS 31–60

RAG & Agent Pipelines

Deploy production vector indexes and workflow copilots.

DAYS 61–90

Team Enablement & Scale

Establish Copilot mentorship and telemetry governance.

PANEL DISCUSSION

Ready for Technical Deep Dive

Prepared to discuss RAG architecture, agent orchestration, Copilot developer workflows, and banking security.

Role: AI Engineer

Programme: Morgan Stanley / TCS

Location: London (3 Days On-Site)